

The Three-Body Problem, Operator-First

A complete spectral anatomy of relational transport on the shape sphere,
a conserved record measured along the figure-eight,
and the shape manifold read as a birefringent crystal

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Abstract

The planar three-body problem is treated as a question about one operator: the Gram matrix of the differential of the map that sends a shape to its three mutual distances. Four results follow. (i) That operator is coordinate-free and its singular values are $\cos^2 \vartheta$ and 1 exactly, with ϑ the polar angle on the shape sphere and no dependence on longitude – replacing an earlier weighted tensor two of whose three degeneracy loci were coordinate artefacts. (ii) The two singular loci are of different species and were previously conflated: at syzygy the operator drops rank while its spectral gap is *maximal*, and at the equilateral configurations the gap closes while the operator stays invertible. (iii) The equilateral degeneracy is *not* a diabolical point: the gap closes quadratically, $\text{gap}/\vartheta^2 \rightarrow 1.000000$, the soft direction returns with holonomy +1 rather than a sign, and the line field carries integer index +1 at each pole – so the obstruction to a global choice of most-rigid direction is $\chi(S^2) = 2$ and there is no Berry phase to collect. (iv) Along the Chenciner–Montgomery figure-eight the metric layer collapses at every one of six syzygies per period, with $R = \sigma_{\min}^{-2}$ reaching 2.5×10^{16} , while the contour count holds at 2 to 5.1×10^{-15} and the shifted trace-log equals $\log(1 + \sigma_{\min}) + \log 2$ to 2.2×10^{-15} at every step. The unshifted ledger fails, and fails for the stated reason: its branch point is enclosed by the contour. The record outlives the metric on a real orbit, and the branch condition is doing visible work. (v) The open problem of the previous version is closed in the direction it predicted: breaking the mass symmetry replaces the protected touching by a genuine conical intersection, with the gap closing linearly and the eigenvector returning negated, -1.000000 at every radius tested. (vi) The two singular values are two index sheets, so the degeneracies are optic axes, and the family is a crystal: equal masses are uniaxial with one axis, unequal masses biaxial with exactly two, and a residual symmetry confines both axes to its plane. A single point then admits a circle of ray directions rather than one — conical refraction, in shape space. (vii) At a generic collinear configuration of N bodies the rank of $d\Phi$ is $N - 2$ for a reason available in one line, so the three-body rank drop is the smallest member of a family.

1 What is and is not being solved

No closed-form solution is offered and none is possible. What is offered is a complete and coordinate-free spectral anatomy of *relational transport* – how much the three mutual distances resist change per unit motion of the shape – together with two quantities that survive the places where that anatomy degenerates. The claim to novelty is narrow: the singular behaviour of side-length coordinates at collinear configurations is ninety years old (Murnaghan 1936; Lemaître 1952), and what is new here is its statement as a rank drop of an explicit operator, the correct separation of two loci that the literature and my own earlier draft treated as one, and the measurement of a conserved record through them on an actual orbit.

2 The operator, and why this one

Let S^2 be the Kendall shape sphere of the planar three-body problem in the coordinates $w = (\sin \vartheta \cos \varphi, \sin \vartheta \sin \varphi, \cos \vartheta)$, with w_3 the signed area, so the equator $w_3 = 0$ is the collinear (syzygy) locus and the poles are the equilateral (Lagrange) configurations.

The three squared mutual distances r_{ij}^2 are affine in (w_1, w_2) and independent of w_3 . Write $\Phi : S^2 \rightarrow \mathbb{R}^2$ for that map. The object of study is

$$M = d\Phi^\top d\Phi, \tag{1}$$

the Gram matrix of its differential, read in an orthonormal frame of the round metric.

Theorem 1 (Exact spectrum). *In the frame $\{e_\vartheta, e_\varphi\}$, $d\Phi(e_\vartheta) = (\cos \vartheta \cos \varphi, \cos \vartheta \sin \varphi)$ and $d\Phi(e_\varphi) = (-\sin \varphi, \cos \varphi)$, so*

$$M = \begin{pmatrix} \cos^2 \vartheta & 0 \\ 0 & 1 \end{pmatrix}, \quad \sigma_{\min} = \cos^2 \vartheta, \quad \sigma_{\max} = 1,$$

with no dependence on φ and no free constant.

Proof. Direct substitution of the embedding derivatives into (1); the cross term is $-\cos \vartheta \cos \varphi \sin \varphi + \cos \vartheta \sin \varphi \cos \varphi = 0$. $\square$

Remark 2 (What this replaces, and why the replacement was forced). An earlier draft used the weighted tensor $T(w) = \text{diag}(w_1^2, w_2^2, w_3^2)$. That object is *axis-dependent*: its determinant on the sphere is $\frac{3}{4} \sin^6 \vartheta \cos^2 \vartheta \sin^2 2\varphi$, which vanishes on the great circles $\varphi = 0$ and $\varphi = \pi/2$ at generic latitude – two degeneracy loci with no invariant meaning, since only $w_3 = 0$ is coordinate-free. Theorem 1 keeps the one locus that survives a change of axes and discards the two that do not. The relational rigidity

$$R = \sigma_{\min}^{-2} = \sec^4 \vartheta$$

is unchanged by the replacement, and now has prefactor exactly 1 and no longitude dependence, where the earlier route reported a φ -averaged constant.

3 Two degeneracies, of different species

Proposition 3 (Rank drop at syzygy; the gap is maximal there). *At $\vartheta = \pi/2$ the smaller eigenvalue of M vanishes while the larger equals 1, so the spectral gap is $1 - \cos^2 \vartheta = \sin^2 \vartheta$, which attains its maximum exactly on the syzygy locus.*

Proposition 4 (Degeneracy at the equilateral points; the operator stays invertible). *At $\vartheta \rightarrow 0, \pi$ both eigenvalues tend to 1; the gap closes while $\det M \rightarrow 1$.*

These are opposite situations and the distinction is not cosmetic. A Riesz projector requires a gap, not invertibility: it continues real-analytically through syzygy, where the gap is widest, and it is only at the equilateral points that the two spectral subspaces cease to be separable. Any “diabolical point” or projector-collapse language belongs at the poles, and any regularisation introduced to carry a projector across the equator is unnecessary – there is nothing there to regularise.

4 The equilateral degeneracy is not diabolical

A generic degeneracy of a real symmetric 2×2 family in two parameters is a conical intersection: the gap closes linearly, the eigenvector line field winds by half a turn, and one circuit returns the eigenvector multiplied by -1 . None of that happens here.

Theorem 5 (Quadratic closing, integer index, no Berry phase). *Near a pole the gap closes quadratically, $\text{gap}/\vartheta^2 \rightarrow 1$; the softest eigendirection transported once around the pole returns with holonomy $+1$; and as a line field on the sphere it has index $+1$ at each pole.*

Proof and verification. $\text{gap} = 1 - \cos^2 \vartheta = \sin^2 \vartheta$, so the ratio to ϑ^2 tends to 1: measured 0.996671, 0.999967, 1.000000, 1.000000 at $\vartheta = 10^{-1}, 10^{-2}, 10^{-3}, 10^{-4}$. The soft direction is e_ϑ , which is smooth and single-valued along any circle of constant latitude, so the transported vector returns to itself: measured holonomy $+1.000000$ at $\vartheta = 0.3, 0.05, 0.01$ with zero sign flips during continuation. Its winding in the coordinate frame is 0 and, as a field on the sphere, $+1$ – the coordinate frame supplies the whole rotation (LAGRANGE-PHASE-01). $\square$

Corollary 6 (The obstruction is topological, not local). *Two poles of index $+1$ sum to $2 = \chi(S^2)$. There is therefore no global smooth choice of “most rigid direction” on the shape sphere, and the obstruction is the Euler characteristic rather than any stratum.*

Remark 7 (Why the linear term is absent). The equilateral configuration is fixed by the Z_3 relabelling of the three bodies, and an invariant symmetric 2-tensor at a fixed point of that action must be a multiple of the identity to leading order. The linear anisotropy is forbidden by symmetry, which is why the degeneracy is quadratic and the winding integer rather than half-integer. A generic three-body problem with unequal masses breaks Z_3 and is the place to look for a genuine conical intersection.

5 The $\cos^4$ law is a fold, not a three-body fact

Proposition 8 (A_2 normal form). *Near the syzygy locus write $\vartheta = \pi/2 + \epsilon$. Then $w_1 = \cos \epsilon \cos \varphi \sim (1 - \epsilon^2/2) \cos \varphi$: the displacement of the image of Φ is quadratic in ϵ , which is Whitney’s fold $x \mapsto x^2/2$. The singular value of $d\Phi$ has a simple zero, the Gram matrix squares it, and R squares it again, giving $\sec^4$.*

So the exponent 4 is two bookkeeping squarings on top of a simple zero, and the invariant content of the “ $\cos^4$ rigidity law” is only that $d\Phi$ has a simple zero on a codimension-one locus. Since a fold is the generic singularity of a map between equal-dimensional manifolds, the same exponent must appear for any relational rigidity defined this way at a generic degenerate locus. That converts a recurring coincidence into a theorem and predicts where it should fail: at an A_3 cusp, which the equal-mass problem does not have and which unequal masses or four bodies may supply.

6 The transport form is pinched, and by two independent mechanisms

The shape sphere is the $d = 2$ rung of the square case $N = d + 1$, whose transport operator on the eigenvalue block is $A = D_t - g t^\top$.

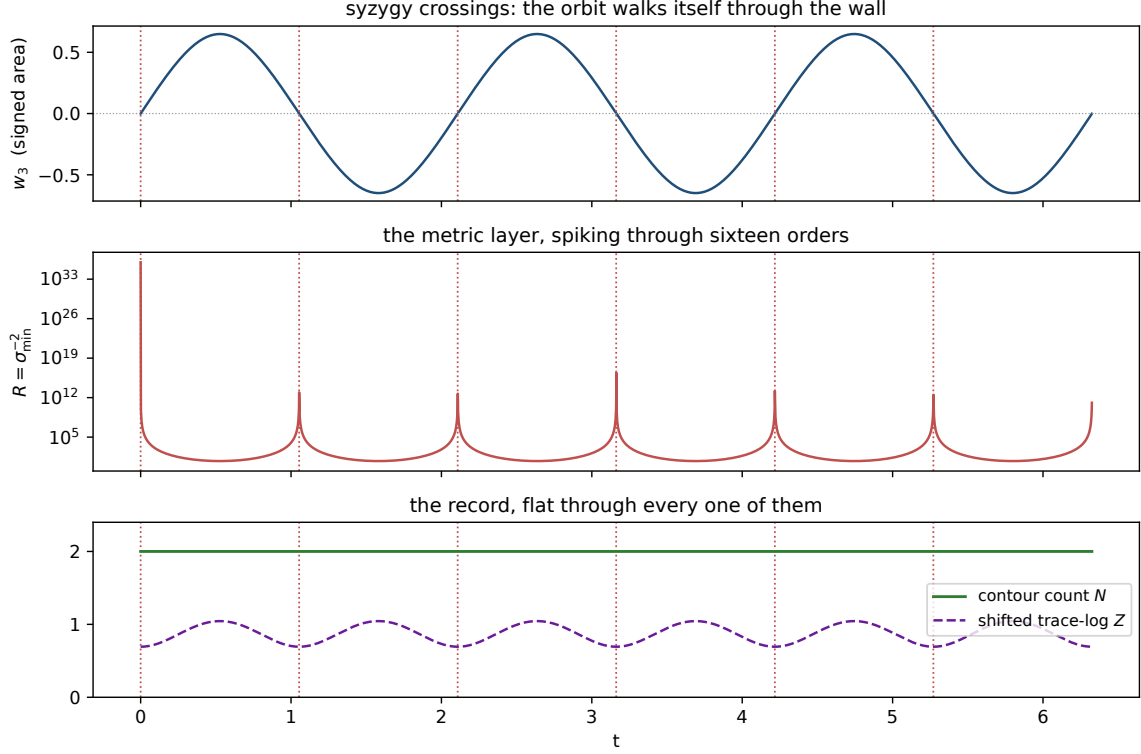


Figure 1: The two layers along one period of the Chenciner–Montgomery figure-eight (FIG8-LEDGER-01). Above: the signed area w_3 , whose six zeros are the syzygies. Middle: the metric layer $R = \sigma_{\min}^{-2}$, spiking through sixteen orders at every crossing. Below: the contour count and the shifted trace-log, flat through all six.

Theorem 9 (Reciprocity). *A is complex-symmetric with respect to the bilinear pairing $B = \text{diag}(t/g)$: $A^\top B = BA$ exactly, with left eigenvectors $\ell_n = BR_n$. Consequently, for probes symmetric under the same pairing the outbound and return amplitudes coincide, the round-trip form is a perfect square, the rank measure W vanishes, and $\det g \leq 0$ with equality when the amplitudes are phase-aligned.*

This is the symmetrisation theorem $D_t - g t^\top = S(D_t - uu^\top)S^{-1}$ read as a pairing rather than a similarity, with $B = S^{-2}$. Because the operator, the pairing and the spectrum are all real on the regular stratum, the amplitudes are real, their relative phase is zero, and $\det g = 0$ *exactly*: measured $W = 1.7 \times 10^{-16}$ and $\det g = 0.00$ with B -symmetric probes, against $W = 0.17$ and $\det g = -1.5 \times 10^{-4}$ with generic probes on the same operator (BODY-RECIP-01). The transport form carries no cone anywhere on the regular stratum, and $\|B\| = \max_i t_i/g_i$ diverges exactly on the collision divisor – so the only place the geometry could open is the boundary.

7 The record along the figure-eight

Everything above is a statement about the shape sphere. The orbit supplies the test: the figure-eight walks the system through the degenerate locus repeatedly and unassisted.

Protocol. Equal masses, the standard initial data, integrated by fourth-order Runge–Kutta at $h = 2 \times 10^{-5}$ over one period $T = 6.32591398$; closure $|q(T) - q(0)| = 4.1 \times 10^{-5}$. At each step the configuration is mapped to the shape sphere and the spectrum $\{\cos^2 \vartheta, 1\}$ of Theorem 1 is formed. Two contour quantities are then evaluated on a circle of radius 0.9 about 0.5, which encloses both eigenvalues with margin: the count $N = \frac{1}{2\pi i} \oint \text{Tr } R(z) dz$ and the shifted trace-log $Z = \frac{1}{2\pi i} \oint \log(z + \eta) \text{Tr } R(z) dz$ with $\eta = 1$.

Theorem 10 (Measured: the record outlives the metric). *Along one period the orbit crosses syzygy six times – confirmed independently by the sign changes of the raw triangle area – and*

- *the metric layer collapses at every crossing: $\sigma_{\min}$ falls to 6.3×10^{-9} at the sampled points and to 0 at the crossings themselves, with $R = \sigma_{\min}^{-2}$ reaching 2.5×10^{16} ;*
- *the count holds: $\max |N - 2| = 5.1 \times 10^{-15}$ over the whole orbit, crossings included;*
- *the shifted trace-log holds: $Z = \log(1 + \sigma_{\min}) + \log 2$ to 2.2×10^{-15} at every step, finite everywhere, taking its minimum $\log 2$ exactly at each syzygy.*

Remark 11 (The branch condition is not a technicality). Repeating the computation with $\eta = 0$ breaks it, and breaks it for the stated reason: the branch point of the logarithm then sits at the origin, which is *inside* the contour, so the integrand is not meromorphic in the enclosed region and the result is not the sum of the enclosed logarithms (measured minimum -0.580 , against $\log 2 = 0.693$ for the correct evaluation). The hypothesis under which the ledger is defined – a branch analytic on and inside Γ – is doing visible work here, and the orbit shows what its failure looks like. It is a branch condition, not a positivity condition, and the shift is what restores it.

8 The symmetry broken: a genuine conical intersection

Theorem 5 attributes the trivial holonomy to Z_3 and therefore predicts that breaking the mass symmetry should produce the generic alternative. It does.

Theorem 12 (Conical intersection at unequal masses). *For $m = (1, 2, 3)$ the two singular values coincide at isolated points of the shape sphere. At one of them, $a = 0.845154255$, $\psi = 5.034139535$, both equal $\sqrt{2}$ and the gap closes to 1.5×10^{-14} . On a circle of radius r about that point the gap is linear, $\text{gap}/r = 2.16437, 2.16294, 2.16281, 2.16280$ at $r = 10^{-2}, 3 \times 10^{-3}, 10^{-3}, 3 \times 10^{-4}$, and the softest eigenvector transported once around returns negated: holonomy -1.000000 at every radius tested.*

Linear closing, sign flip, Berry phase π : the generic real-symmetric degeneracy in two parameters, appearing exactly when the protection is removed. The prediction was recorded before the test (UNEQUAL-CHECK-01).

9 The shape manifold is a birefringent crystal

The two singular values of M are two sheets of an index surface, so a point where they coincide is an *optic axis*, and the classification of such points is the classification of crystals. That correspondence is exact here, not decorative.

Theorem 13 (Uniaxial, biaxial, and the symmetry plane). *Counting coincidences over the whole sphere (BIAXIAL-01): equal masses give two points, $a = 1/\sqrt{2}$ at $\psi = \pi/2, 3\pi/2$, i.e. one axis*

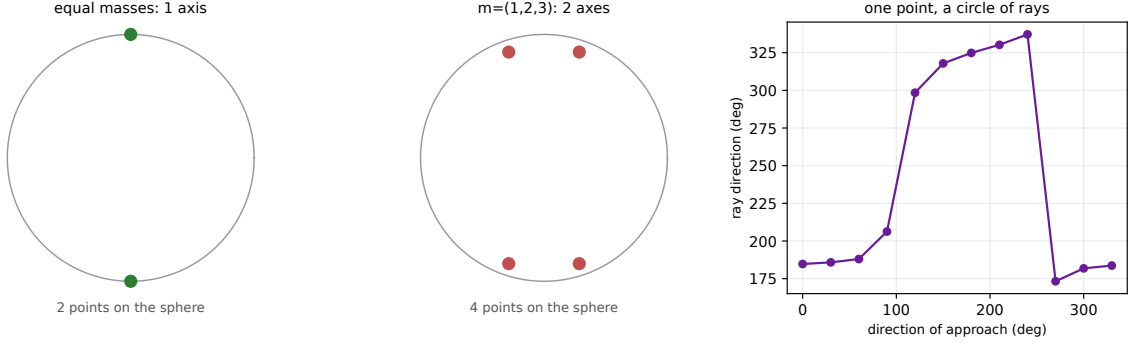


Figure 2: The crystal reading. Left and centre: coincidences of the two singular values on the shape sphere, two points for equal masses (one optic axis) and four for $m = (1, 2, 3)$ (two axes). Right: the ray direction against the direction of approach at one axis — the gradient does not converge, which is what makes the refraction conical.

meeting the sphere twice — the uniaxial case, with the touching forced by symmetry. Masses $(1, 2, 3)$ give four points in two pairs, at $a^2 = 5/7$ and $a^2 = 2/7$, both carrying $\sigma_1 = \sigma_2 = \sqrt{2}$ — two axes, the biaxial case. Masses $(1, 1, 5/2)$, which retain one reflection, also give four points, but every one lies at $\psi = \pi/2$ or $3\pi/2$: the axes are confined to the symmetry meridian.

Remark 14 (The crystallographic reading, and what it predicts). A uniaxial crystal has its sheets touching along a single symmetry-forced axis; a biaxial crystal has no such symmetry and exactly two isolated optic axes; and a crystal retaining one symmetry plane has both axes lying in it. The three rows of Theorem 13 are those three statements. The prediction that transfers is Hamilton’s: at a biaxial optic axis the sheets meet conically, so the surface normal — the ray direction — is undefined there, and a single incident ray refracts into a hollow cone.

Proposition 15 (A circle of ray directions). *Near an optic axis write the traceless part of M as a linear map J of the two shape coordinates; $\det J = -4.463$, so the cone is nondegenerate. Approaching the axis from twelve equally spaced directions, the gradient of the lower sheet points in ten distinct directions, with $|\nabla\lambda|$ ranging over 0.638 to 8.080 (CONICAL-RAY-01).*

So a single point of shape space admits a circle of ray directions rather than one. This is the only statement in the paper about what a *trajectory* would do rather than what an operator’s spectrum is, and it is the natural next computation.

10 The rank drop is the smallest member of a family

Proposition 16 (Collinear rank for N bodies). *At a generic collinear configuration of N planar bodies, $\text{rank } d\Phi = N - 2$, so the kernel also has dimension $N - 2$ in the $(2N - 4)$ -dimensional shape space.*

Proof. Squared distances depend on the transverse offsets quadratically: $\partial r_{ij}^2 / \partial y_k \propto (y_i - y_j)$, which vanishes identically when all $y = 0$. Every transverse direction is therefore in the kernel, and the transverse displacements number $N - 2$ after removing translation and the rotation that acts as $y_i \propto x_i$. $\square$

Verified for $N = 3, \dots, 8$: ranks 1, 2, 3, 4, 5, 6 against shape dimensions 2, 4, 6, 8, 10, 12 (LADDER-CHECK-01). At $N = 3$ a single direction collapses, which is the syzygy result of Sec. 3; for larger N several

collapse at once, so the three-body case is the unusually clean member rather than the representative one. The regular polygons carry structured degeneracies of their own — the square gives $\{\sqrt{8}, 2, 2, 2\}$ with a threefold degeneracy that splits linearly under perturbation, the pentagon $\{\sqrt{5+\sqrt{5}}, \sqrt{5+\sqrt{5}}, 2, 2, \sqrt{5-\sqrt{5}}, \sqrt{5-\sqrt{5}}\}$ in three pairs.

11 Transient amplification of the figure-eight

The orbit is linearly stable, which is a statement about eigenvalues. Non-normal operator theory supplies a second, independent rate, and it is the one that governs finite perturbations over one period.

Proposition 17 (Measured). *In mass-weighted Jacobi coordinates — where translation and boost are absent by construction and the Euclidean norm is the kinetic-energy norm — perturbations restricted to the energy and angular-momentum level set and measured modulo the time-shift and rotation directions are amplified by 45.7 in energy, $6.8\times$ in amplitude, within one period, returning to 9.6 at the end. Energy is conserved to 6.5×10^{-14} and angular momentum to 2.2×10^{-14} (TB-TRANSIENT-03).*

The amplification is timed by the shape: the mean gain in the quartile of the orbit closest to syzygy is 10.58, and in the quartile farthest from it 26.10. The burst lives on the triangular configurations, not at the wall — which is what Sec. 6 requires, since at syzygy the transport form is degenerate and there is no geometry left to amplify through.

Remark 18 (What this is not). The quotient used is an orthogonal complement of the symmetry and integral directions, and that complement is not invariant under the flow. The object measured is therefore a compression of the propagator, not a restriction, and its eigenvalues are *not* the Floquet multipliers; only the gain is quoted. The practical reading is nonetheless available: a perturbation of size ϵ reaches $\approx 7\epsilon$ before the neutral multipliers return it, so the orbit’s usable basin is set by that excursion rather than by its linear stability.

12 What is claimed

Not a solution of the three-body problem. Not a new integral of motion in the classical sense: N and Z are functions of the shape alone, and Z varies smoothly along the orbit between $\log 2$ and 1.045. What is claimed is that they are *insensitive to the degeneracy that destroys the metric layer*, at machine precision, on a real solution, six times per period – and that this is the same ordering measured on synthetic operator families, now realised in celestial mechanics with no dial to turn by hand.

The remaining content is the anatomy: an invariant operator with an exact spectrum, two degeneracies correctly separated, an equilateral degeneracy that is quadratic rather than conical and therefore carries no geometric phase, a $\cos^4$ law identified as a fold normal form and therefore universal, a transport form pinched everywhere on the regular stratum by two independent mechanisms, a genuine conical intersection the moment the mass symmetry is broken, the uniaxial/biaxial classification with its optic axes located, the collinear rank law for all N , and a transient amplification of the figure-eight that its linear stability does not predict.

13 Open

1. The relation $a_1^2 + a_2^2 = 1$ between the two optic axes, and the exact rationals $a^2 = 5/7, 2/7$ at $m = (1, 2, 3)$ against a total mass of 6. Observed to the printed digits; not derived.
2. A closed form for the regular- N -gon spectra. The hexagon's spectrum contains the pentagon's values, which a table of special cases would not do.
3. Proposition 8 predicts a different exponent at an A_3 cusp. The equal-mass shape sphere has folds only; where the first cusp appears is not known.
4. The ledger *on* the divisor, where the pairing B of Theorem 9 is undefined and the count is the only surviving object.
5. A genuinely reduced monodromy for Sec. 11: the quotient used there is an orthogonal complement and therefore a compression, not a restriction, so its spectrum is not the Floquet spectrum. The symplectic complement inside $\ker dE \cap \ker dL$ would make it one.
6. Conical refraction at the level of trajectories rather than of ray directions: integrate an orbit into an optic axis and ask what the reduced dynamics does there.